

Noken Rademacher Descent

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Abstract

Semantic refinement of Noken attention, via query-key similarity alignment, is achieved by Rademacher descent of implicit model hyperparameters.

Introduction

An implicit deep learning model consists of a pair of equations, equilibrium and prediction. Next-token and seq-to-seq contextual token learning can be modeled as such; the equilibrium equation relates values and queries, it's fixed point, called a Noken, is a latent counterpart to the input token (from a context string), and the prediction equation relates the keys corresponding to values and queries. Provided the value matrix is well-posed for the activation function, we can jointly learn the key-value and key-query pairs of embedding matrices and the Noken. Noken implicit models can be composed into chains and feedback loops, creating a knowledge graph, edges added when Renyi α -entropy is reduced.

Noken implicit models take a particular form where the prediction equation is parameterized by a constant, that we get to choose. We now outline a Rademacher entropy descent algorithm with step size $\eta > 0$ and number of iterations $N > 1$. Generic Noken implicit models are of the form:

$$\begin{aligned} s_i &= \varphi(Vs_i + Qt_i) \text{ [equilibrium]} \\ \varepsilon &= K^*s_i + Kt_i \text{ [prediction]} \end{aligned}$$

where input t_i is a target token from a context string $t_1, t_2, \dots, t_m$, and V is well-posed for φ to ensure that unique fixed points exist, and hyperparameter ε is a constant real column vector.

Start with vector $\varepsilon = 0$, and compute the resulting average Renyi α -entropy E_0 of the token contextual representations with the keys that give semantics to queries. Next draw $\varepsilon = \{\sigma_l\}$, where σ_l are independent with $\text{Prob}(\sigma_l = \eta) = 1/2 = \text{Prob}(\sigma_l = -\eta)$ and once again compute the resulting average Renyi α -entropy E_ε for this new implicit model. Select the fixed point with the least E , suppose it's E_ε , now draw ε' and compare E_ε and $E_{\varepsilon'}$, once again choose the least, and repeat this perturbation process N times. The Rademacher draws of ε , descend to a Noken implicit model with potentially lower average Renyi α -entropy. This process refines the semantic alignment of queries and keys. So besides configuring Noken network composition chains and feedback loops, we can also utilize stochastic choice of Noken implicit model hyperparameter ε , as a means to reduce entropy and strengthen attention meaning.

For token string $t_1, \dots, t_m$ learn matrices

$V, Q, \mathcal{N}, K$ and fixed point state s_i

so that for $i \in [m]$:

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = \mathcal{N}s_i + Kt_i$$

where V is well-posed for φ

and ε is a constant vector.

Let $\alpha_{ij} = \text{softmax score}(q_i, x_j)$ where score is a similarity function and $q_i = Qt_i, x_j = \mathcal{N}s_j$

Let $h_i = \sum_{j=1}^m \alpha_{ij} v_j$, where $v_j = Vs_j$

and $\beta_{ij} = \text{softmax score}(h_i, k_j)$, $k_j = Kt_j$

Let $E_\varepsilon = E\{\beta_{ij}\} = \frac{1}{m} \sum_{i=1}^m H_\alpha^i$, $H_\alpha^i = H_\alpha(\{\beta_{ij}\}_{j=1}^m)$

and H_α is Renyi α -entropy.

Rademacher Descent:

Hyperparameters: step size η

of iterations N

Initial step

Set $S = 0 \in \mathbb{R}^n$

Draw $\varepsilon_1 = (\sigma_1, \dots, \sigma_n)$ where σ_i are independent

$$\text{Prob}(\sigma_i = S + \eta) = \frac{1}{2} = \text{Prob}(\sigma_i = S - \eta)$$

Induction step

Given E_k , set $S = E_k$ and

draw $\varepsilon' = (\sigma'_1, \dots, \sigma'_n)$ where σ'_i are independent

$$\text{Prob}(\sigma'_i = S + \eta) = \frac{1}{2} = \text{Prob}(\sigma'_i = S - \eta)$$

$$\text{Let } E_{k+1} = \begin{cases} \varepsilon' & \text{if } E_{\varepsilon'} < E_{E_k} \\ E_k & \text{ow} \end{cases}$$

Repeat N times.

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